

Process-oriented Business Performance Management with Petri Nets

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Abstract

Modern business performance management (BPM) comprises modeling, analyzing and monitoring of business processes. A variety of different approaches for business performance management have emerged recently. One issue common to these approaches is the inability to link the business process schema to the information gathered during the relevant phases. In this paper, we present a Petri net-based approach for process-oriented business performance management. Our approach is based on a new type of high-level Petri nets, so called XML nets. In XML nets business process objects, performance indicators and the performance measurement system are represented with XML. Being a consistent and comprehensible methodology for the whole process life cycle, XML nets support integrated process-oriented business performance management. We will also demonstrate the modules and basic functions of a Petri net-based software prototype designed to support process-oriented BPM.

1. Introduction

A variety of commercial software tools and academic methodologies relating to Business Performance Management (BPM) have emerged during the last years [5, 7]. One issue common to these approaches is the inability to link the business process model to the information gathered during the relevant phases of BPM. In particular, the integration of performance indicator information in all phases of BPM is missing. Thus, an adequate business process model should allow for defining both the specific process goals and the corresponding performance

indicators and then for using the current performance indicator values to coordinate and control the business process execution. Process-oriented BPM can be divided into three interdependent phases: process modeling, analyzing and monitoring. For an efficient BPM the integration of performance indicator information in all phases is necessary. In particular, specific target values or tolerance ranges have to be considered while simulating process scenarios. Therefore, adequate constructs for modeling performance indicators, target values and tolerance ranges have to be defined.

The paper is structured as follows: After a description of related work and the comparison to the suggested methodology, an introduction to business process modeling with XML nets will be presented. Thereafter, XML net-based modeling concepts of performance indicators are outlined. We describe the enhancements of the XML net-based business process analysis and monitoring. Thereupon we will demonstrate the basic functions of a new software prototype, called INCOME2010. The paper concludes with a summary of advantages of the presented approach and a brief outlook on future research and implementation work.

2. Related Work

Different methods and description languages have been proposed for process modeling, most of which are based on textual programming languages or graphical notations. Semi-formal methods are very popular, widely disseminated and easy to use. They do, however, exhibit major shortcomings especially in regard to supporting all phases of business performance management with one integrated method. On the basis of different definitions and interpretations of their

respective static and dynamic modeling components different types of Petri nets have been derived.

Nested Relation/Transition nets for example combine the concept of complex structured objects with the Petri net formalism [11]. Motivated by the increasing acceptance and dissemination of the XML standard [13], XML nets were derived (for a detailed description we refer to [8, 9]). XML nets are a graphical modeling language for document based processes and especially for the manipulation of XML documents.

[14] highlights the necessity of employing a software tool for integrated modeling, analysis and control of manufacturing systems. In [1] a toolkit is described for standard business processes that can be graphically combined in order to model logistics or manufacturing systems. A variety of well-defined performance measurement systems is proposed, most notably the balanced scorecard [6].

All these approaches make a clear distinction between the process modeling and the monitoring of performance indicators.

3. Process-oriented BPM with XML nets

Petri nets [12] are a formal graphical process description language that combines the advantages of the visual representation of processes with a formal definition. Petri nets consist of static components (places, depicted by circles) and dynamic components (transitions, depicted by rectangles) that are connected by edges.

In XML nets the static components are typed by XML schema diagrams, each of them representing an XML schema. Places can thus be interpreted as containers for XML documents representing relevant process objects. The flow of XML documents is defined by the firing of transitions, i.e., the execution of activities. A transition may be inscribed by a logical expression of the variables that appear in the labels of the adjacent edges. The expression evaluates either to true or false for an instantiation of the variables. So-called filter schemas label the adjacent edges and thereby describe the process objects that are relevant for the activity and the way these objects are manipulated (created, changed or deleted).

3.1. Modeling Performance Indicators with XML nets

Performance indicators map the business circumstances onto quantitative values (e.g., system status, time value, cost, etc.). They are mainly used to

ascertain the economic efficiency of the business as well as to determine the management objectives and to verify the implementation of those objectives (i.e., monitoring variance between current and target). From a process-oriented perspective, performance indicators can be interpreted as the input or the output of (several) activities.

In XML nets performance indicators are represented by XML schemas. Thus, places that are typed by a performance indicator schema (as a part of the overall schema for the performance measurement system) are interpreted as containers for the performance indicator value. The calculation of the performance indicator value can be modeled by parts of an XML net. We distinguish the following types of performance indicators:

- *Counter*: a value that may be increased or decreased (e.g., number of rejected components).
- *Difference*: a value that results from the difference of two other values (e.g., working capital).
- *Average*: a value that represents an average over a certain period (e.g., average of rejected orders weekly).
- *Partition*: a value that represents the ratio of a part to the totality (e.g., rejection ratio daily).
- *Indexes*: a value that results from the ratio of two values, typically used to explain statistical series (e.g., labor cost index).
- *Correlation (relative)*: a value representing the ratio of two numerical values of different concepts (e.g., return on equity).

Each type of performance indicator can be modeled by a specific XML net pattern (for a general concept of workflow patterns see for example [2]).

An *average* performance indicator depends on a specific, but fixed time interval. The respective XML net pattern is modeled as a transition with two input places and one output place. One place contains the start date and start time for the relevant interval, the other one the performance indicator. Figure 1 shows a pattern for a weekly aggregation. In this case seven performance indicator values are read concurrently. Moreover, the transition inscription ensures that the values of seven subsequent days are taken into account.

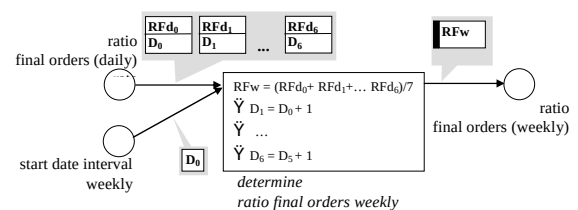


Figure. 1. XML net for an average pattern

For a more flexible aggregation of performance indicator values (for example based on time intervals with arbitrary start and end) a different pattern has to be applied: the calculation of the average cannot be modeled as one single transition (with the calculation rule as inscription), but as a subnet that models the calculation by iterations.

Patterns for *counter*, *indexes*, *differences*, *partitions* and *correlations* can be derived in a similar manner to the presented average pattern.

3.2. Modeling Alert and Repair Workflows with XML nets

Performance indicator violations, i.e., the deviation of a performance indicator value from its pre-defined target value or tolerance range, can be interpreted as an exceptional state of the business process. For the description of performance indicator violations we propose two different types of transitions, so-called *alert* and *repair* transitions.

Alert transitions (depicted by a rectangle with a double line on the left side) are related to the concept of fact transitions [3] and express requirements on performance indicator values. In case of being enabled, the alert transition indicates a state where the respective requirement is violated. The alert is logged and, for example, the sending of an email to the responsible person can be triggered. In comparison to alert transitions, repair transitions (depicted by a rectangle with a triangle on the left side) trigger a repair workflow which is linked to the respective business process. For example, the raw material reliability (RMR) is one critical performance indicator and determines the performance of a complete inter-organizational manufacturing process. When an assembler signals that an involved raw material distributor has an actual raw material reliability below the target value (RMRT) the repair transition "repair material reliability" is enabled (Figure 2).

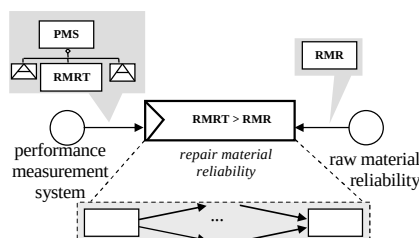


Figure 2. Repair transition

For each repair transition, a corresponding repair pattern must exist: an XML net refining the transition is

assigned to each repair transition. In our example a repair workflow could comprise the selection and authorization of an alternative raw material distributor. The repair workflow is triggered as soon as the repair transition is enabled.

3.3. Business Performance Analysis by Simulation

Analysis in process-oriented BPM aims, for example, at validating the defined exceptional states and the corresponding alerts and repair workflows. For the simulation of XML nets with alert and repair transitions two different simulation strategies are applicable: either exceptional states modeled by alert and repair transitions can be reached or not. In the first case, simulation provides information concerning activities which have to be accomplished such that business goals are achieved. In the second case, the simulation reveals exceptional states (and a set of activities which led to these states). For this reason, repair workflows are also simulated and therefore subject to validation.

Simulation can, e.g., be used to evaluate a business process schema with different target values for a certain performance indicator. For this purpose, process times, represented by firing delays and stochastic process outputs, can be assigned to the transitions. Simulation of XML nets allows for business process analysis on a very detailed level. For instance, it is possible to parameterize simulation runs and change target values. Moreover, different alerts or repair workflows for a performance indicator can be simulated. Thus, simulation allows for exploring different exceptional states and deploying the appropriate reaction before implementing the business processes.

4. The INCOME2010 Software Prototype

The software prototype INCOME2010 provides fully integrated modules for Petri net-based modeling, analysis and monitoring. The ObjectSchemaEditor is used for modeling (the structure of) the relevant objects as graphical XML schemas. Whereas the ObjectSchemaEditor is used for data modeling, the NetModeler is employed for generating the process schema. A high-level Petri net (e.g. an XML net) can be generated: The basic Petri net schema is combined with the graphical XML schemas, defined with the ObjectSchemaEditor. The system ensures consistency by only providing feasible specification alternatives for the manipulation filter. The user can optionally define

transition inscriptions on the basis of a pre-defined set of logical operators. For representation purposes, different views on the modelled Petri net schema can be selected. For example, manipulation filters or transition inscriptions can optionally be blanked out. The previously modelled Petri net can be analysed with the PerformanceSimulator. The system automatically generates a simulation schema, performs a pre-determined number of simulation runs and stores the simulation results. With the module PerformanceMonitor the operational data is linked to the process, modeled with the NetModeler. For monitoring purposes, the module provides pre-configured patterns for the definition of all relevant performance indicators, which can be integrated into the process schemas. Furthermore the PerformanceMonitor has a special mode for the definition of the performance measurement system. Due to its open architecture with configurable interfaces, the tool can directly be linked to internal and external systems, e.g. data warehouse, ERP and other applications in order to exchange relevant process data.

5. Conclusion and Outlook

A major issue of process-oriented BPM is the appropriate representation of performance indicators, target values and performance measurement systems. The inability to model performance indicators and their corresponding performance measurement system in an adequate way severely limits the performance indicator based simulation and monitoring. In this paper we have presented a Petri net-based approach for process-oriented BPM which addresses this essential issue. For the modeling of performance indicators, alert and repair workflows performance indicator patterns have been presented. Petri nets facilitate an integrated approach to business performance management by using a consistent and comprehensible method throughout all relevant phases.

The INCOME2010 software prototype currently under development can be directly linked to the transactional systems in an enterprise. The prototype constitutes the basis for further development, implementation and application of the proposed method. For example, the pattern based PerformanceMonitor will be extended in order to support the flexible derivation of new, user defined patterns. Currently we have adopted the presented approach among others to supply chain management [10] and workflow-based learning-object management [4].

6. References

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